

Assignment 1 Solutions

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October 10, 2025

Problem 1

Given the reads

\$ATGC, ATGC, TGCAT, ATGCT, GCTAG, CTAGC, TAGCA, AGCA\$, ATGCA, GCATG, TGCTA, CATGC,

construct the de Bruijn graph with *4-mers as nodes* (so edges are 5-mers) and output the assembled sequence.

Method (concise). With $k = 4$ -mer nodes, each observed 5-mer becomes a directed edge from its 4-mer prefix to its 4-mer suffix. The contig is recovered by an Eulerian path that traverses each edge once; spelling the path yields the assembled sequence.

Step 1: 5-mers (edges). Sliding a window of length 5 over all reads and deduplicating yields the set of edges:

$\{ \$ATGC, TGCAT, ATGCT, GCTAG, CTAGC, TAGCA, AGCA$, ATGCA, GCATG, TGCTA, CATGC \}$.

Step 2: Edge mapping (prefix $\rightarrow$ suffix).

\$ATGC : \$ATG $\rightarrow$ ATGC TGCAT : TGCA $\rightarrow$ GCAT ATGCT : ATGC $\rightarrow$ TGCT GCTAG : GCTA $\rightarrow$ CTAG
CTAGC : CTAG $\rightarrow$ TAGC TAGCA : TAGC $\rightarrow$ AGCA AGCA\$: AGCA $\rightarrow$ GCA\$ ATGCA : ATGC $\rightarrow$ TGCA
GCATG : GCAT $\rightarrow$ CATG TGCTA : TGCT $\rightarrow$ GCTA CATGC : CATG $\rightarrow$ ATGC

Step 3: Degree check and Eulerian path. All nodes are balanced except:

\$ATG : out = 1, in = 0 (start), GCA\$: in = 1, out = 0 (end).

Hence an Eulerian path exists from \$ATG to GCA\$. One valid traversal is:

\$ATG $\rightarrow$ ATGC $\rightarrow$ TGCA $\rightarrow$ GCAT $\rightarrow$ CATG $\rightarrow$ ATGC $\rightarrow$ TGCT $\rightarrow$ GCTA $\rightarrow$ CTAG $\rightarrow$ TAGC $\rightarrow$ AGCA $\rightarrow$ GCA\$

Step 4: Assembled sequence (spell the path). Start with the first 4-mer \$ATG; each step appends the last character of the next node. The result is

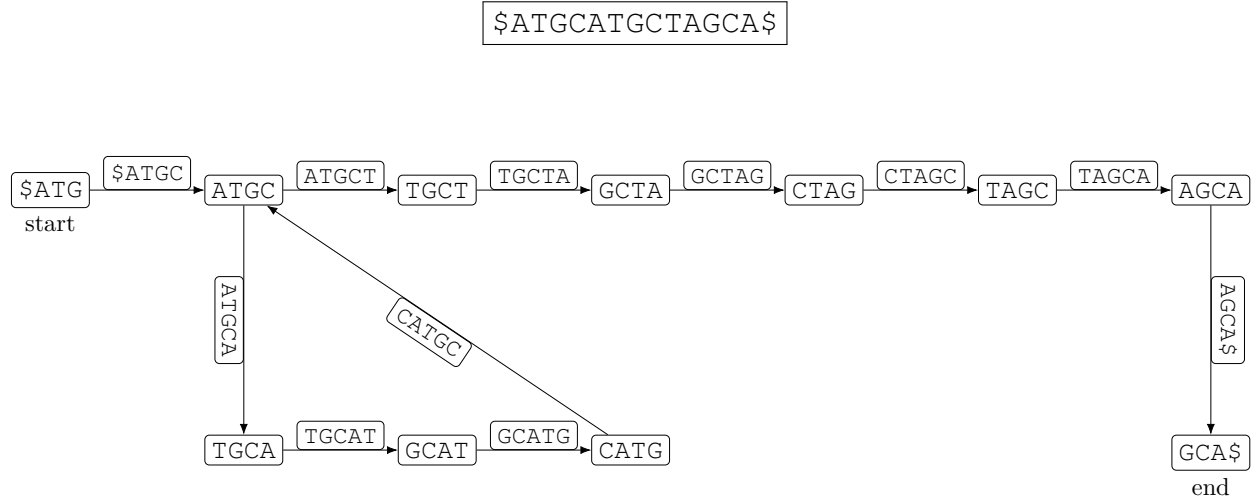


Figure: de Bruijn graph with 4-mer nodes and 5-mer edge labels.

Problem 2

In an abstract sequence of length 18, we are given candidate exons (closed intervals) with weights:

$$[1, 4] : 4, \quad [3, 6] : 5, \quad [3, 8] : 10, \quad [9, 10] : 1, \quad [8, 16] : 12, \quad [15, 18] : 6, \quad [13, 16] : 7, \quad [7, 11] : 8.$$

Find a maximum-weight subset of *non-overlapping* exons. We treat intervals as overlapping if they share at least one position; thus two chosen intervals $[L_i, R_i]$ and $[L_j, R_j]$ must satisfy $R_i < L_j$ or $R_j < L_i$.

Model. This is the classic *Weighted Interval Scheduling* (WIS) problem. Let each exon j have left endpoint L_j , right endpoint R_j , and weight w_j . Sort intervals by nondecreasing R . For each interval j , define the *compatibility index*

$$p(j) = \max\{i < j : R_i < L_j\},$$

with $p(j) = 0$ if no such i exists. The optimal value $M[j]$ for the first j intervals satisfies the recurrence

$$M[0] = 0, \quad M[j] = \max(M[j-1], w_j + M[p(j)]) \quad (j \geq 1).$$

We reconstruct an optimal set by taking interval j exactly when $w_j + M[p(j)] > M[j-1]$.

Data preparation. Label the intervals for convenience:

$$\begin{aligned} A &= [1, 4] : 4, & B &= [3, 6] : 5, & C &= [3, 8] : 10, & D &= [9, 10] : 1, \\ E &= [8, 16] : 12, & F &= [15, 18] : 6, & G &= [13, 16] : 7, & H &= [7, 11] : 8. \end{aligned}$$

Sort by R :

$$A(1, 4, 4), B(3, 6, 5), C(3, 8, 10), D(9, 10, 1), H(7, 11, 8), E(8, 16, 12), G(13, 16, 7), F(15, 18, 6).$$

Compute $p(j)$ using the strict non-overlap condition $R_i < L_j$:

$$p(1) = 0, p(2) = 0, p(3) = 0, p(4) = 3, p(5) = 2, p(6) = 2, p(7) = 5, p(8) = 5.$$

Dynamic programming table. Let w_j denote the weight of the j -th interval in the sorted list. The DP values are:

j	Interval (L, R, w)	$p(j)$	$w_j + M[p(j)]$	$M[j - 1]$	$M[j] = \max$
0	—	—	—	—	0
1	(1, 4, 4)	0	4	0	4
2	(3, 6, 5)	0	5	4	5
3	(3, 8, 10)	0	10	5	10
4	(9, 10, 1)	3	$1 + 10 = 11$	10	11
5	(7, 11, 8)	2	$8 + 5 = 13$	11	13
6	(8, 16, 12)	2	$12 + 5 = 17$	13	17
7	(13, 16, 7)	5	$7 + 13 = 20$	17	20
8	(15, 18, 6)	5	$6 + 13 = 19$	20	20

Thus the optimal value is $M[8] = \boxed{20}$.

Answer. A maximum-weight non-overlapping set is

$$\boxed{\{[3, 6], [7, 11], [13, 16]\}}$$

with total weight

$$\boxed{5 + 8 + 7 = 20}.$$

Problem 3

Align

$$\text{Sequence_A} = \text{GATTCATA} \quad (\text{length } 8), \quad \text{Sequence_B} = \text{GGATCGA} \quad (\text{length } 7),$$

with scoring: match = +5, mismatch = -2, indel (gap) = -6. Construct the DP matrix and report the optimal (global) alignment score.

Algorithm (global alignment). Let $A_1 \dots A_n$ and $B_1 \dots B_m$ be the two sequences ($n = 8$, $m = 7$). Define the score matrix $D \in \mathbb{Z}^{(n+1) \times (m+1)}$:

$$D(0,0) = 0, \quad D(i,0) = i \cdot (-6), \quad D(0,j) = j \cdot (-6),$$

$$D(i,j) = \max \left\{ \underbrace{D(i-1,j-1) + s(A_i, B_j)}_{\text{diag (match/mismatch)}}, \underbrace{D(i-1,j) + (-6)}_{\text{up (gap in B)}}, \underbrace{D(i,j-1) + (-6)}_{\text{left (gap in A)}} \right\},$$

where $s(x,y) = +5$ if $x = y$, otherwise -2 .

Dynamic programming matrix.

	-	G	G	A	T	C	G	A
-	0	-6	-12	-18	-24	-30	-36	-42
G	-6	5	-1	-7	-13	-19	-25	-31
A	-12	-1	3	4	-2	-8	-14	-20
T	-18	-7	-3	1	9	3	-3	-9
T	-24	-13	-9	-5	6	7	1	-5
C	-30	-19	-15	-11	0	11	5	-1
A	-36	-25	-21	-10	-6	5	9	10
T	-42	-31	-27	-16	-5	-1	3	7
A	-48	-37	-33	-22	-11	-7	-3	8

Optimal score. The optimal global alignment score is

$$\boxed{8}.$$

One optimal alignment (traceback). A valid traceback yielding the score 8 is:

$$\begin{array}{cccccccc} G & A & T & T & C & A & T & A \\ G & G & A & T & C & - & G & A \end{array} \Rightarrow 5 - 2 - 2 + 5 + 5 - 6 - 2 + 5 = 8.$$

Problem 4

Results. The search recovered high-confidence matches in *Ambystoma mexicanum* and lower-similarity hits annotated as titin-like homologs in *Procambarus clarkii*. The salamander sequence showed the strongest similarity ($E = 0.0$; high coverage), consistent with a conserved mismatch-repair core. In contrast, crayfish titin-like sequences had modest identity despite significant E-values, suggesting distant homology or domain-level similarity. The MSA highlighted conserved motifs in the ATPase/endonuclease-associated regions typical of MutL family proteins, whereas low-complexity regions were variable.

Table 1: Selected MLH1 Homologs Identified by BLASTp

Accession	Description	Species	Score	Length	Identity	E-value	Coverage	Positives	Query ID
XP_069502281.1	trichohyalin-like	<i>A. mexicanum</i>	665	8003	97%	0.0	26.69%	3374	XP_069502281.1
XP_069493420.1	trichohyalin-like	<i>A. mexicanum</i>	169	1053	94%	1e-36	43.85%	2865	XP_069493420.1
XP_069173201.1	titin homolog X10	<i>P. clarkii</i>	104	603	41%	6e-17	44.25%	8612	XP_069173201.1
XP_069173198.1	titin homolog X7	<i>P. clarkii</i>	104	603	41%	7e-17	44.25%	8742	XP_069173198.1
XP_069173196.1	titin homolog X5	<i>P. clarkii</i>	104	603	41%	7e-17	44.25%	8761	XP_069173196.1
XP_069173193.1	titin homolog X2	<i>P. clarkii</i>	104	603	41%	7e-17	44.25%	8771	XP_069173193.1
XP_069173194.1	titin homolog X3	<i>P. clarkii</i>	104	603	41%	7e-17	44.25%	8771	XP_069173194.1
XP_069173192.1	titin homolog X1	<i>P. clarkii</i>	104	603	41%	7e-17	44.25%	8772	XP_069173192.1
XP_069173197.1	titin homolog X6	<i>P. clarkii</i>	104	603	41%	7e-17	44.25%	8756	XP_069173197.1
XP_069173199.1	titin homolog X8	<i>P. clarkii</i>	104	603	41%	7e-17	44.25%	8737	XP_069173199.1
XP_069173195.1	titin homolog X4	<i>P. clarkii</i>	104	601	41%	8e-17	44.25%	8766	XP_069173195.1
XP_069173200.1	titin homolog X9	<i>P. clarkii</i>	104	601	41%	8e-17	44.25%	8728	XP_069173200.1
XP_069173203.1	titin homolog X12	<i>P. clarkii</i>	104	601	41%	8e-17	44.25%	8497	XP_069173203.1
XP_069173202.1	titin homolog X11	<i>P. clarkii</i>	103	598	41%	1e-16	44.25%	8599	XP_069173202.1
XP_069173204.1	titin homolog X13	<i>P. clarkii</i>	101	583	41%	7e-16	44.25%	7316	XP_069173204.1

Phylogeny and Interpretation. The tree resolved *P. clarkii* sequences as a tight clade, reflecting recent duplication/isoforms, while *A. mexicanum* grouped more closely to vertebrate-like MLH1. Branch lengths indicated stronger conservation within vertebrates and greater divergence in invertebrate titin-like proteins.

Conclusion. BLASTp, MSA, and phylogenetic inference collectively indicate that *MLH1* is evolutionarily conserved in vertebrates, with more pronounced divergence among invertebrate homologs.

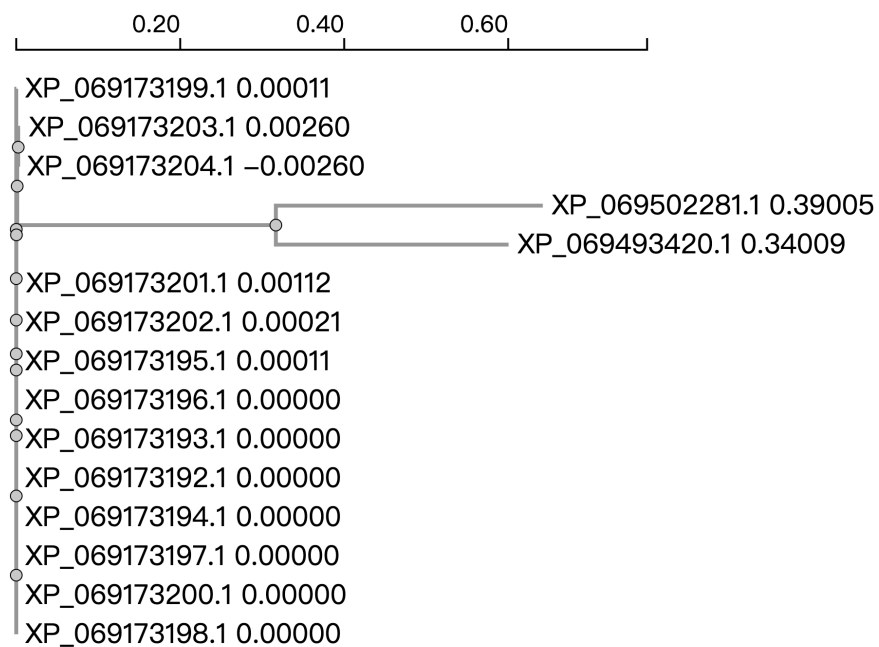


Figure 1: Phylogenetic Tree